AI - Lab

2. Datasets & Dataloaders



Hochschule Karlsruhe
Technik und Wirtschaft
UNIVERSITY OF APPLIED SCIENCES

Dr. Fabian Sigloch

WS 26/27

Datasets

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- In the last exercise, we used the existing `torchvision.datasets.MNIST` dataset and passed it to a data loader:

```
train_loader = torch.utils.data.DataLoader(
    torchvision.datasets.MNIST(
        './files/', train=True, download=True,
        transform=torchvision.transforms.Compose([
            torchvision.transforms.ToTensor(),
            torchvision.transforms.Normalize((0.1307,), (0.3081,))
        ])
    ),
    batch_size=batch_size_train, shuffle=True
)
```

- You can also write your own `Dataset` classes to load custom data.

Datasets

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- A tutorial on writing custom datasets, dataloaders, and transforms can be found here.

WRITING CUSTOM DATASETS, DATALOADERS AND TRANSFORMS

Author: Sasank Chilamkurthy

A lot of effort in solving any machine learning problem goes into preparing the data. PyTorch provides many tools to make data loading easy and hopefully, to make your code more readable. In this tutorial, we will see how to load and preprocess/augment data from a non trivial dataset.

To run this tutorial, please make sure the following packages are installed:

- `scikit-image`: For image io and transforms
- `pandas`: For easier csv parsing

```
from __future__ import print_function, division
import os
import torch
```

Assignment 2

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Task:

1. Write a custom dataset class for the Titanic data (in the data folder on GitHub). Use only `Pclass`, `Age`, `SibSp`, `Parch`, `Fare`, `Sex`, and `Embarked`. Preprocess the features in this class (scaling, one-hot encoding, ...) and split the data into training and validation data (80% / 20%). Its constructor should be usable as:

```
titanic_train = TitanicDataSet('titanic.csv', train=True)
titanic_val = TitanicDataSet('titanic.csv', train=False)
```

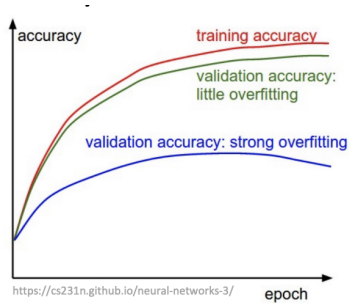
2. Build a neural network with one hidden layer of size 3 to predict passenger survival. Use binary cross-entropy (BCE) loss; the output layer needs a sigmoid activation. Train with a shuffled data loader and batches of size 16.
3. Evaluate validation performance using accuracy.

Assignment 2

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Task:

4. Create the plot shown on the right, introduced in the lecture.
5. Increase network complexity with additional layers and neurons. Can you overfit the training data?
6. Try to reduce overfitting by introducing a dropout layer.



Submission Guidelines

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- Send your solution as a Jupyter notebook (`.ipynb`) by e-mail. Include your team name and the names of all team members.
- Use the subject: `[KI-Lab] Assignment 2 - <teamname>`
- Deadline: next Thursday, 11:59 p.m. (23:59).

Final words

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- You can start working on Assignment 2 now.
- You and your partners can organise your time as you see fit.
- I will be available for questions until the end of the slot at 1 p.m.
- Questions can also be asked on Mattermost. Please do not post solutions there.

- Any more questions?
- Happy coding!